CFRM 502, Winter 2020.

Homework 3

Due: Monday Febraury 3rd, 2020, NO LATER than 11:59pm

Instructions: You must submit your work through Canvas, and in the form of two attachments: A .pdf file of your detailed answers and a file containing your codes (.R or .Rmd). The .pdf files can be typeset or be a readable scan of your handwritten notes. You do not need to include R outputs in the .pdf file. However, you must clearly indicate the relevant part of your R codes when answering a question that is based on your code.

The maximum number of points you can receive for this homework is 15.
Question 4 will not be graded.

1. (5 pts) Consider problem 3 of the previous homework about forward rate estimation. The zero-coupon data can be loaded from "HW3-Q1.txt" that is posted with this homework. Use the following code

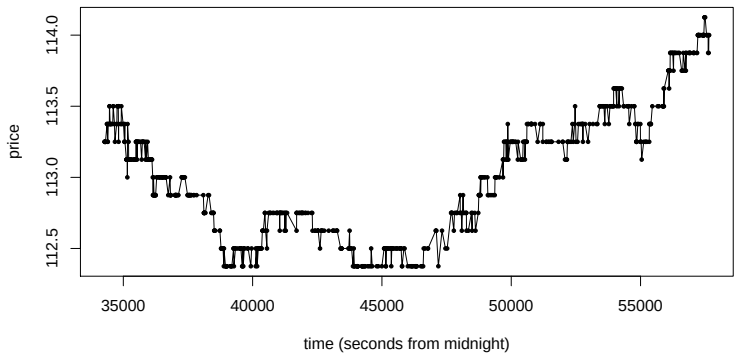
```
dat = read.table("HW3-Q1.txt",header=TRUE)
dat = dat[order(dat$T),]
EMP = -diff(log(dat$price))/diff(dat$T)
MAT = dat$T[-1]
```

   In the previous Homework, you have fitted the following model.

$$\text{EMP}_i = \beta_0 + \beta_1 \,\text{MAT}_i + \beta_2 \,\text{MAT}_i^2 + \beta_3 \,\text{MAT}_i^3 + \varepsilon_i.$$

   In question 3.(e) of Homework 2, you should have found that the residuals of the model above do not have constant variance. Suggest a model to remedy this abnormality. Fit the suggested model and confirm that the issue is resolved (by repeating the regression diagnostic).

2. (5 pts) Consider the intraday trading of IBM stock on November 21, 1990, as shown in the graph below. There were 726 transactions on that day during normal trading hours, but only 195 trades resulted in a price change.



   Let t_i be the time of the i-th trade that resulted in a price change. The "duration" $\Delta t_i = t_i - t_{i-1}$ is the time duration between price changes. Furthermore, let N_i be the number of trades in the time interval (t_{i-1}, t_i) that resulted in no price change.

   The values of $\{\Delta t_i\}$ and $\{N_i\}$ are included in the file "HW3-Q2.dat". To load the data, use the following R code:

```
dat = read.table("HW3-Q2.dat",header=TRUE)
DUR = dat[,"duration"]
N = dat[,'trades']
```

which stores the values of $\{\Delta t_i\}$ and $\{N_i\}$ in variables "DUR" and "N", respectively.

(a) We believe that the conditional probability $\mathbb{P}(N_i = 0|\Delta t_i)$ is a function of $\log(\Delta t_i)$. So, we propose the following model:

$$\mathbb{P}(N_i = 0|\Delta t_i) = \alpha_0 + \alpha_1 \log(\Delta t_i),$$

for some unknown constants α_0 and α_1. Estimate α_0 and α_1 using OLS (ordinary least squares).

(b) There is a problem with the model in part (a). Explain the problem. Then, suggest a model to remedy the problem and fit it.

3. (5 pts) The data file "HW3-Q3.txt" to be used for this problem contains 1259 rows giving the daily adjusted closing values of the S&P 500 index during the period beginning January 4, 2008 and ending December 31, 2012. The goal of this problem is to produce a forecasting for the future changes in the index. We shall use the first 4 years for training of our prediction system and we shall test them on the last year.

Use the following code to load the data

```
SP = read.table("HW3-Q3.txt",header=TRUE)[,2]
```

Let $\{S_t\}$ denote the index values.

(a) (2 pts) Construct a local polynomial regression using observations 61-1000 of the index to predict the next weekly change in S&P 500 using the changes over the last day, the last week, the last 3 weeks and the last 12 weeks as predictors. Assume that every 5 observations is 1 week. Thus, use $S_{t+5} - S_t$ as your response variable and the values $S_t - S_{t-1}$, $S_t - S_{t-5}$, $S_t - S_{t-15}$, and $S_t - S_{t-60}$ as predictors.

(b) (1 pts) Use the local polynomial regression model estimated in part (a) to predict 254 values of the 5 day increment in the index for the last year of data. That is, predict $S_{t+5} - S_t$ for $t \in 1001, \ldots, 1254$ based on the model fitted in part (a). Compute the sum of squared errors in this predictions.

Hint for parts (a) and (b): As mentioned during the lecture, the "locpoly()" function cannot handle multiple regression (i.e. more than one predictors). Instead, use the "loess()" function. Assuming that "Dat.train" and "Dat.test" are data frames with columns Y, X1, X2, X3, and X4, use

```
fit=loess(Y~X1+X2+X3+X4, Dat.train,
          control = loess.control(surface = "direct"))
```

for fitting the model. For prediction, use

```
Yh = predict(fit, Dat.test)
```

(c) (2 pts) Assume that in part (b), instead of using the local polynomial regression, you directly use the last week's change in S&P 500 index as the prediction for its next weekly change. Find the sum of squared errors of this trivial predictor. Which predictor is better, the one in part (a) or the trivial predictor? Interpret the result.

4. (This question will not be graded) Solve the following exercises from the textbook, section 21.8.

   (a) Exercise 3. There is a typo in the definition of $s(x)$. Use the following definition: $s(x) = (x)_+ - 2(x-1)_+ + (x-2)_+$.

   (b) Exercise 4.